

Reasonable Doubt and Appellate Review Through Probabilistic Causal DAGs

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Abstract

Natural-language verdicts cannot reliably satisfy the reasonable doubt standard when many causal variables are at issue. The graph-theoretic conditions for ruling out alternative explanations are non-obvious to unaided verbal reasoning. We propose formalizing legal proof through probabilistic causal directed acyclic graphs (DAGs) grounded in Pearl’s structural causal model framework, addressing the Dawid-Faigman-Fienberg criticism through a focus on appellate review of judgment-weight consistency. Reasonable doubt is defined as an active alternative causal path the evidence has not d-separated; the beyond-reasonable-doubt threshold is anchored in the Probability of Necessity. Criminal proceedings impose a path-closure obligation on the prosecution, while civil proceedings require a path-weighting competition between competing DAGs - structurally distinct tasks carrying different obligations. We further formalize Wright’s NESS test within structural causal models and provide an edge-weight elicitation protocol grounding every weight in traceable evidentiary passages.

Keywords: causal DAG; reasonable doubt; legal causation; d-separation; criminal procedure; appellate review

1 Introduction

1.1 The Problem

In criminal jurisprudence, establishing causality is not merely a binary exercise. It is a probabilistic one constrained by the burden of proof beyond a reasonable doubt. A verdict must not only assert a causal link between the defendant’s action and the harm: it must implicitly dismantle all other plausible explanations. As the US Supreme Court recognized in *In re Winship*, 397 U.S. 358 (1970), the reasonable doubt standard is a constitutional bedrock, and the same holds in most countries.

Yet judicial opinions routinely obscure the weight of competing narratives. A judge may state that “the evidence overwhelmingly points to the defendant” without formally defining the weight of that evidence or specifying which alternative explanations have been considered and foreclosed. This opacity creates two systemic risks. First, appellate courts lack a principled framework for auditing whether a factfinder genuinely examined and refuted alternative causal explanations. Second, asymmetries in legal representation mean that the structural adequacy of causal reasoning goes unchallenged in the majority of cases.

Cameron Todd Willingham was executed in Texas on February 17, 2004 for arson murder. Nine independent fire scientists - in three separate reviews spanning 2004 to 2009 - subsequently concluded that every forensic indicator of arson relied upon at trial was scientifically unsupported, and that an accidental ignition path had never been excluded Hurst (2004); Carpenter et al. (2006); Beyler (2009). The Texas Forensic Science Commission acknowledged unreliable fire science in its final report Texas Forensic Science Commission (2011). The question the present framework asks is not whether Willingham was guilty or innocent; it is whether the verdict’s causal reasoning was structurally adequate - whether the alternative causal path was d-separated by the evidence the factfinder actually relied upon. It was not.

This failure is not unique to Willingham. It is structural: natural-language verdicts

cannot reliably perform the graph-theoretic operations that the reasonable doubt standard implicitly requires.

1.2 The Core Argument

Judea Pearl has emphasized that the necessity of formal graphical representations becomes apparent precisely when the number of variables and causal pathways is large enough that informal, intuitive reasoning about dependence and independence systematically fails (Pearl, 2009). A court proceeding is exactly such a context. A moderately complex criminal case - six plausible causal scenarios, five categories of physical evidence, four witnesses, three contested mechanisms - generates a graph whose d-separation properties cannot be assessed by unaided intuition. Research on conditional independence judgment shows that human reasoners systematically confuse conditioning on a mediator (which blocks a path) with conditioning on a collider (which opens one), and that these errors persist even among trained experts when arguments are presented verbally rather than graphically (Pearl, 2009).

Legal verdicts are presented verbally. When a judge writes that “the pre-existing cardiac condition does not negate the defendant’s liability,” the sentence conflates two structurally distinct graph operations: the eggshell-plaintiff doctrine extending liability once but-for causation is established, and the separate question of whether but-for causation has in fact been established against the pre-existing condition as an alternative sufficient cause. Only a formal graph makes the conflation visible. The Supreme Court’s unanimous reversal in *Burridge v. United States* (Supreme Court of the United States, 2014) reached the same conclusion by doctrinal reasoning; the present framework would have surfaced the same structural deficiency computationally from the trial record itself.

The argument for causal DAGs is therefore not merely one of convenience or precision. It is that natural-language causal reasoning in complex cases is cognitively unreliable in ways that are systematic, predictable, and correctable by a formal graph. When there are many variables and evidence nodes to consider, a formal graph is a must that makes the reasonable

doubt standard enforceable rather than rhetorical.

1.3 The Electoral Integrity Context

The stakes of structural opacity extend beyond individual criminal proceedings. In South Korea, the Supreme Court ruled twice - in Cases No. 2020Su5028 (2022) and 2022Su5053 (2024) - that statistical evidence of early-voting anomalies did not satisfy the evidentiary standard for election nullification, because plaintiffs failed to identify the perpetrator, timing, and method of any alleged violation (Supreme Court of Korea, 2022, 2024). This standard, which we show is the verbal expression of a graph-theoretic requirement to name all nodes on an asserted causal path, left the question of whether alternative causal paths had been adequately closed without a formal framework for resolution.

Recent forensic and regression-based analysis of four electoral cycles from 2020 to 2025 (Kim, 2026) demonstrates that 73-94% of geographic variation in early Democratic vote shares is explained by observable sociodemographic characteristics, with coefficients nearly identical to those for same-day votes. In DAG terms, the demographic-sorting alternative path was empirically d-separated; the manipulation path’s Probability of Necessity does not reach any plausible criminal threshold. Yet no formal framework existed by which the court could verify that this closure was structurally complete rather than merely asserted. We return to this case in Section 7, where we propose three procedural principles for the adjudication of electoral integrity claims under the DAG framework.

1.4 Relationship to Prior Work

The present contribution is positioned at the intersection of three bodies of literature that have not previously been integrated.

Kadane and Schum (Kadane and Schum, 1996) applied Bayesian networks to legal argument structure, but their networks operate at the observational layer of Pearl’s inferential hierarchy and cannot represent interventional or counterfactual quantities. They cannot cer-

tify that an alternative causal path has been closed; they can only report that it has a lower posterior probability than the prosecution’s hypothesis. These are different claims.

Halpern (Halpern, 2016) provides a rigorous structural-equation treatment of actual causation, handling preemption and overdetermination through his modified HP definition. But Halpern’s framework presupposes *known* structural equations and determines causation as a binary property. It cannot represent epistemic uncertainty over which causal edges exist, it does not map causation onto graduated proof standards, and it does not address the verdict audit operation - the question of whether an existing verdict’s stated reasoning is structurally adequate.

Pearl and Tian-Pearl (Pearl, 2009; Tian and Pearl, 2000) define the Probability of Necessity and derive sharp bounds on it, connecting counterfactual causation to estimable population quantities. The legal connection is noted but undeveloped: the mapping of PN onto the hierarchy of legal proof standards, and the application of sharp bounds to evidentiary edge weights rather than population data, are contributions of the present framework.

Section 2 develops these comparisons in full, including a component-by-component table.

1.5 Contributions and Organization

The paper makes four contributions:

1. **A formal theory of reasonable doubt as a DAG property** (Section 3): the definition of a Reasonable Doubt Path; the Reasonable Doubt Condition (Proposition 1); the NESS formalization within SCMs; the Civil/Criminal Standard Duality (Proposition 2); and the Weight Elicitation Protocol (Definition 8).
2. **The graph-complexity argument for DAG adoption** (Section 4): the case that formal graphs are necessary, not merely convenient, in complex causal adjudication; the computational structure of d-separation and path enumeration; and the structural asymmetry between criminal and civil proof standards as different graph-theoretic

tasks.

3. **Four real-case applications** (Section 5): *Willingham* (arson, structural deficiency confirmed post-execution), *Burrage/Cadden* (overdetermination in drug-death prosecutions, reversal and conviction respectively), and *Kennedy* (novus actus interveniens), along with the South Korean election integrity trials in Section 7.
4. **A framework for electoral integrity adjudication** (Section 7): grounded in the Korean Supreme Court proceedings, the forensic evidence from four electoral cycles, and three procedural principles for the graph-based evaluation of electoral fraud claims.

The remainder of the paper is organized as follows. Section 2 surveys and differentiates the prior literature. Section 3 develops the formal mathematical framework. Section 4 argues for the necessity of formal graphs and analyses the computational and structural properties of the civil/criminal asymmetry. Section 5 presents the four case studies. Section 6 addresses implementation, weight elicitation, and jurisdictional constraints, including a note on automated extraction as a future direction. Section 7 develops the electoral integrity application. Section 8 concludes.

2 Related Work

Table 1 summarizes the relationship between the present framework and the four bodies of prior work most directly relevant to its formal claims. The subsections below develop each comparison in detail.

2.1 Causation in Legal Doctrine

The dominant doctrinal tools - “but-for” causation and proximate cause - were designed for natural-language reasoning rather than formal computation.

Table 1: Comparison of the Present Framework with Principal Prior Works

Component	K&S (1996)	Halpern (2016)	Pearl (2009)	Tian– Pearl	This work
Causal (interventional) layer	No	Yes	Yes	Yes	Yes
Counterfactual PN/PS defined	No	No	Yes	Yes	Yes
Epistemic weights on causal edges	Partial ^a	No	No	No	Yes
Proof standards as PN thresholds	No	No	No	No	Yes
D-sep of alt. paths as RD closure	No	No	No	No	Yes
NESS within SCMs	No	Substantially	No	No	Extension ^b
Verdict audit operation	No	No	No	No	Yes
Civil/criminal standard duality	No	No	No	No	Yes

^a K&S assign Bayesian priors to *evidence items*, not to the existence of causal edges in a structural model.

^b Our contribution is the connection of NESS to evidentiary thresholds under contested structural equations, not the NESS-within-SCM mapping itself.

But-for causation. The but-for test asks: would the harm have occurred but for the defendant’s act? Formally, but-for causation holds when $P(H \mid \neg D) \approx 0$. This is precisely the counterfactual operation $P(H_{D=0})$ in Pearl’s potential outcomes framework (Pearl, 2009), making it amenable to DAG representation. However, the but-for test famously fails in cases of causal overdetermination (multiple sufficient causes), as *Burridge v. United States*, 571 U.S. 204 (2014) illustrates in the drug-death context (Supreme Court of the United States, 2014).

The NESS test. Wright’s Necessary Element of a Sufficient Set (NESS) test (Wright, 1985) generalizes but-for causation: a condition is a cause of a consequence if it is a necessary element of a set of existing antecedent conditions that is together sufficient to produce the consequence. NESS maps naturally onto the notion of a minimal sufficient cause set in structural causal models, which we formalize in Section 3.

Probabilistic causation in law. Scholarship by Kaye (Kaye, 1982), Tribe (Tribe, 1971), and Allen and Pardo (Allen and Pardo, 2019) has explored whether formal probability theory can anchor legal proof standards. Our framework extends this line of inquiry by grounding evidentiary sufficiency directly in causal - rather than merely statistical - structure, and by making the distinction between the two structurally precise.

2.2 Kadane and Schum (1996): The Limits of Evidential Bayesian Networks

The application of Bayesian networks to legal reasoning was pioneered by Wigmore’s “chart method” (Wigmore, 1913) and later formalized by Kadane and Schum (Kadane and Schum, 1996) in their exhaustive probabilistic reconstruction of the Sacco and Vanzetti trial record. Their framework is sophisticated and represents the most complete prior attempt to bring formal probability to legal argument structure.

However, Kadane and Schum’s networks operate entirely within the *observational* layer of Pearl’s inferential hierarchy (Pearl, 2009). Every quantity they compute is of the form $P(H \mid E_1, E_2, \dots, E_k)$ - the probability of guilt conditional on the observed evidence. Their framework cannot represent, and therefore cannot reason about:

1. *Interventional queries*: $P(H \mid do(D))$ - what would happen if we intervened to set the defendant’s act, rather than merely observed that it occurred. Proximate cause reasoning requires this layer.
2. *Counterfactual queries*: $P(H_{D=0} = 0 \mid D = 1, H = 1)$ - the probability of necessity, which but-for causation requires. A Bayesian network can tell you that the presence of an accelerant lamp is probabilistically associated with accidental fire; it cannot tell you whether the defendant’s act was *necessary* for the fire given the lamp’s presence, because necessity is a counterfactual quantity, not a conditional one.
3. *D-separation as evidentiary closure*: Kadane and Schum treat competing hypotheses as alternatives to be assigned prior probabilities and updated via Bayes’ rule. The present framework makes a fundamentally different claim: an alternative causal path is not merely a hypothesis with lower posterior probability - it is an *active path in the graph* that must be formally d-separated by the introduced evidence before the prosecution’s causal hypothesis can be deemed adequate to the reasonable doubt standard. An alternative path carrying $P(\pi_{alt}) = 0.30$ has a lower posterior than the prosecution’s

hypothesis, yet if it is not d-separated by the evidence record it constitutes a Reasonable Doubt Path under Definition 6, regardless of the posterior comparison. This is a structurally distinct criterion from Bayesian updating, and it maps directly onto the legal intuition that reasonable doubt is not about which hypothesis is more probable, but about whether alternative explanations have been *foreclosed*.

The Willingham case (Section 5.1) illustrates this gap precisely. Kadane and Schum’s framework, applied to *Willingham v. State*, would correctly register a high posterior for the prosecution hypothesis given the fire marshal’s testimony. It could not formally certify that the prosecution’s evidence had failed to close the alternative causal path (accelerant lamp $\rightarrow$ accidental ignition $\rightarrow$ fire), because certifying path closure is a d-separation operation, not a posterior update. Nine independent fire scientists performed that certification post-execution; the present framework would surface the structural deficiency from the trial record itself.

2.3 Halpern (2016): Actual Causality and Its Limits

Halpern’s *Actual Causality* (Halpern, 2016), building on the earlier Halpern-Pearl modified HP definition (Halpern and Pearl, 2001), provides the most rigorous structural-equation treatment of the jury-room causation question in the literature. His framework handles preemption, overdetermination, and goes through a combination of structural equations and contingency analysis, and is explicitly motivated by legal contexts throughout.

The present framework takes Halpern’s foundational contribution as given and extends it in three directions he does not address.

Extension 1: Epistemic uncertainty over structural equations. Halpern’s framework takes the structural equations as *known* (Halpern, 2016). Given the equations and the context $\mathbf{u}$, his modified HP definition determines whether D is an actual cause of H as a *binary* verdict. In a legal proceeding, the structural equations are themselves *contested*: the prosecution asserts one causal model, the defense asserts another, and the factfinder must

adjudicate between them under uncertainty. Our weight function $w : E \rightarrow [0, 1]$ (Definition 3) is an epistemic overlay on the structural equation layer, representing uncertainty over which edges belong in the model at all. This is not a causal strength parameter within a known equation; it is a prior over model structure. Halpern has no analog for this, because his framework presupposes a known model. Remark 1 makes this distinction precise.

Extension 2: Proof standards as thresholds on PN. Halpern determines actual causation as a binary property. Proposition 1 maps the *degree* of causal necessity (PN) onto a graduated threshold that varies by legal proof standard: preponderance of evidence at $\theta > 0.5$, clear and convincing at $\theta > 0.75$, beyond reasonable doubt at $\theta \approx 0.95$. Halpern’s framework answers the causation question; the present framework answers the *evidentiary sufficiency* question - which is a distinct question even when the causal facts are identical. As Proposition 2 demonstrates, this distinction generates a structural asymmetry between criminal and civil proceedings that has no representation in Halpern.

Extension 3: Verdict audit as a distinct operation. Halpern’s framework is a tool for computing actual causation given known facts (Halpern, 2016). The present framework is a tool for *auditing* whether a verdict’s stated reasoning - its asserted edge weights and claimed evidentiary closures - is internally structurally adequate. This is the reframing that resolves the Dawid-Faigman-Fienberg identification objection (Section 2.5): we do not estimate PN from population data; we assess whether the evidence the factfinder actually relied upon supports a PN estimate meeting the applicable threshold. This operation is not in Halpern, and it is not in any prior work in the statistics and law traditions.

On the NESS formalization. We acknowledge that Halpern’s modified HP definition substantially covers NESS-type cases through his contingency-based treatment of overtermination. Our Definition 7 (NESS within SCMs) is not a wholly original contribution relative to Halpern (Halpern, 2016). Our contribution is narrower: the connection of NESS to *evidentiary thresholds under contested structural equations*, particularly in the overtermination cases that arise in drug-death prosecutions (*Burrage*, Section 5.2) where the

competing sufficient sets carry probabilistic weights rather than deterministic truth values.

2.4 Pearl (2009) and Tian-Pearl (2000): PN/PS and Sharp Bounds

Pearl’s structural causal model framework (Pearl, 2009) provides the foundational language of the present work: the do-calculus, the distinction between observational, interventional, and counterfactual quantities, and the definition of PN and PS. Tian and Pearl (Tian and Pearl, 2000) derive sharp bounds on PN when identification from observational data alone fails.

The present framework builds on but extends this body of work in two respects.

Connecting PN to legal proof standards. Pearl and Tian-Pearl develop PN and PS as statistical measures estimable from population data (Tian and Pearl, 2000). The legal connection is noted but not developed: Pearl remarks that PN corresponds to but-for causation, but does not map PN thresholds onto the hierarchy of legal proof standards, nor does he formalize the verdict audit operation. Proposition 1 and the civil/criminal duality of Proposition 2 are contributions at this interface.

Applying sharp bounds to evidentiary edge weights rather than population data. Tian and Pearl’s sharp bounds assume that PN is estimated from observational and experimental data on a population (Tian and Pearl, 2000). The present framework applies structurally identical bounds to *evidentiary edge weights* - quantities derived from the verdict record, not from population sampling. This reapplication sidesteps the identification problem that Dawid, Faigman, and Fienberg rightly flag as irreducible in the trial-court setting (Dawid et al., 2014), because we are not estimating PN from external data; we are bounding it from the internal causal reasoning of the verdict itself. Section 6 makes this reapplication explicit and acknowledges its limitations.

2.5 The Dawid-Faigman-Fienberg Debate and Pearl’s Reply

The most direct prior engagement between Pearlian causal inference and legal epistemology is the published exchange in *Sociological Methods & Research* (2014-2015) (Dawid et al., 2014; Pearl, 2015; Dawid et al., 2015). Dawid, Faigman, and Fienberg distinguished the *effects of causes* (EoC) from the *causes of effects* (CoE), arguing that courts face a CoE question that neither experimental nor observational data can typically answer without untestable monotonicity assumptions. Pearl responded that his PN measure precisely formalizes CoE (Pearl, 2015); Dawid and colleagues replied that individual-level PN is not point-identified from population data in practice (Dawid et al., 2015).

We accept the Dawid-Faigman-Fienberg diagnosis that CoE is the operative legal question, and adopt Pearl’s PN/PS measures as the appropriate formal targets, but we depart from both sides of the debate by operating at the level of *verdict audit* rather than *statistical estimation*. The identification problem is sidestepped, not solved: we bound PN from the verdict’s own evidentiary structure rather than estimating it from external population data. The limitation - that our PN bounds are only as valid as the faithfulness of the evidentiary record - is acknowledged in Section 6.

2.6 AI, LLM, and Automated Legal Reasoning: A Contextual Note

A natural extension of the framework is the automated construction of weighted causal DAGs from verdict text. Large language models and neuro-symbolic pipelines are increasingly capable of extracting causal assertions, hedged language, and dismissed hypotheses from judicial opinions at scale (Chalkidis et al., 2019; Lippi and Torroni, 2016; Hendrycks et al., 2021). Such systems could, in principle, populate the edge weight function $w : E \rightarrow [0, 1]$ directly from textual evidence anchors, with calibrated uncertainty estimates derived via Monte Carlo Dropout (Gal and Ghahramani, 2016) or conformal prediction (Angelopoulos and Bates, 2022).

We treat this as a future direction note rather than a primary claim of the present

paper, for two reasons. First, the formal framework - Definitions 1-7, Propositions 1-2, and the case studies of Section 5 - is fully operative with human-assigned edge weights and does not require an automated extraction pipeline. Second, a defensible automated implementation requires annotated training corpora, calibration benchmarks, and inter-rater reliability studies that lie beyond the scope of the present contribution. We note the direction and leave the implementation to future work.

3 Formalizing Reasonable Doubt in Causal Topology

This section develops the formal mathematical framework. Three of our usages depart from standard practice in the SCM literature and warrant explicit advance notice. First, we overlay an *epistemic weight function* on the edges of the causal DAG (Definition 3); this weight measures evidentiary support for the existence of a given edge, and is distinct from any causal-strength parameter encoded within the structural equation itself. Second, we treat each individual directed causal path from an exogenous source (or unobserved third-party agent) to the harm node as an *independently auditable object* (Definitions 6 and Proposition 1); this is a use of d-separation machinery for verdict audit rather than for the standard task of testing conditional independence between two sets of nodes. Third, we apply Tian and Pearl’s sharp bounds on the Probability of Necessity to *evidentiary edge weights derived from the verdict record* rather than to estimates from population data (Section 3.8); this re-application is what makes the verdict audit operation tractable in the absence of identifying data. Each departure is motivated where introduced below.

3.1 Structural Causal Models: Foundational Definitions

We ground our framework in Pearl’s Structural Causal Models (SCMs) (Pearl, 2009), which provide a unified language for observational, interventional, and counterfactual reasoning.

Definition 1 (Structural Causal Model). A *Structural Causal Model* $\mathcal{M}$ is a 4-tuple $\langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$, where:

- $\mathbf{U} = \{U_1, \dots, U_k\}$ is a set of exogenous (background) variables whose values are determined by factors external to the model;
- $\mathbf{V} = \{V_1, \dots, V_n\}$ is a set of endogenous variables, each determined by other variables in the model;
- $\mathcal{F} = \{f_1, \dots, f_n\}$ is a set of structural equations, where each f_i determines V_i as a function of $\mathbf{Pa}(V_i) \subseteq \mathbf{V} \cup \mathbf{U}$ (the parents of V_i);
- $P(\mathbf{U})$ is a probability distribution over the exogenous variables.

The associated **causal DAG** $\mathcal{G}$ is obtained by drawing a directed edge $V_j \rightarrow V_i$ whenever $V_j \in \mathbf{Pa}(V_i)$.

Definition 2 (Directed Causal Path). A *directed causal path* π in $\mathcal{G}$ is a finite sequence of distinct nodes $V_1 \rightarrow V_2 \rightarrow \dots \rightarrow V_k$ such that each consecutive pair (V_i, V_{i+1}) is a directed edge in $\mathcal{G}$. We call V_1 the source and V_k the sink of π .

For a directed path $\pi : D = V_0 \rightarrow V_1 \rightarrow \dots \rightarrow V_k = H$ from the defendant node to the harm node, we write $P(H = 1 \mid \text{do}(\pi))$ as shorthand for $P(H = 1 \mid \text{do}(D = 1))$ computed under the sub-SCM $\mathcal{M}|_\pi$ in which the structural equations along π operate but no other directed routes from D to H are active. The intervention is on the source node D ; the values of intermediate nodes are determined by the structural equations propagating D 's set value along π . This shorthand notation is used throughout the paper.

Definition 3 (Weighted Causal DAG for Legal Evidence). We augment the standard SCM with a weight function $w : E \rightarrow [0, 1]$ over the edges E of the causal DAG $\mathcal{G}$, where $w(V_j \rightarrow V_i)$ represents the epistemic confidence, derived from the evidentiary record, that V_j belongs to the parent set $\mathbf{Pa}(V_i)$ of V_i in the true SCM of the case. A weight of 1 indicates certainty that the edge $V_j \rightarrow V_i$ is present in the true model; a weight approaching 0 indicates that the

purported edge rests on unreliable or contested evidence. The operationalization of “epistemic confidence” as a calibrated value on a five-point scale grounded in linguistic hedging in the evidentiary record is given in Definition 8.

When weights are aggregated along a path as a product $w(\pi) = \prod_{e \in \pi} w(e)$, this aggregation presupposes path-level independence of the edge-existence judgments. The conditions under which the product approximation is valid, and the marginalization formula required when paths share intermediate nodes, are stated in Remark 4.

Remark 1 (Epistemic weight vs. causal strength). *The weight function $w(V_j \rightarrow V_i) \in [0, 1]$ is an epistemic overlay on the structural causal model: it represents the factfinder’s (or auditing system’s) confidence that V_j belongs to $\mathbf{Pa}(V_i)$ in the true SCM of the case, given the evidence record. It is not a causal-strength parameter within an already-known structural equation. This distinction is consequential in three ways.*

First, it permits the framework to model contested causation under uncertainty: when the structural equations themselves are disputed - the prosecution’s model and the defense’s model differ in their edge sets - the weight function encodes where the dispute lies and how much evidentiary support each contested edge has. Halpern’s actual causality framework (Halpern, 2016) presupposes known structural equations; the weight function is precisely what is needed when equations are unknown and the subject of adversarial contest.

Second, it preserves the causal semantics of the underlying SCM: because w is an overlay on the edge set and not a modification of $\mathcal{F}$, the do-calculus operations remain well-defined on the base model (under any given hypothesised edge configuration), and the PN and PS measures retain their counterfactual interpretation.

Third, it grounds the verdict audit operation: the weight assigned to each edge is traceable to a specific span in the verdict’s evidentiary record (a testimony, a forensic report, a statistical analysis), making the audit report transparent and contestable by appellate review. This traceability is not available in frameworks that treat causal parameters as fixed or estimated from external population data.

3.2 Legal Standards of Proof as Probabilistic Thresholds

Legal proof standards correspond to different constraints on the interventional probability assigned to the prosecution’s causal hypothesis.

Definition 4 (Evidentiary Threshold). *Let π_P denote the prosecution’s directed causal path from defendant’s act D to harm H (cf. Definition 2), and let Π_{alt} denote the set of alternative causal paths in $\mathcal{G}$: directed paths from an exogenous source $U \in \mathbf{U}$ or a third-party agent variable to the harm node H , not passing through the defendant node D . These are the alternative causal accounts that the prosecution must foreclose. Let $\mathbf{E}$ denote the set of evidence variables introduced in the proceeding. The following standards apply:*

$$\text{Preponderance (civil): } P(H = 1 \mid do(\pi_P), \mathbf{E}) > 0.5 \quad (1)$$

$$\text{Clear and convincing: } P(H = 1 \mid do(\pi_P), \mathbf{E}) > 0.75 \quad (2)$$

$$\text{Beyond reasonable doubt (criminal): } P(H = 1 \mid do(\pi_P), \mathbf{E}) > \theta, \quad \theta \approx 0.95 \quad (3)$$

where the threshold θ in (3) is jurisdiction-dependent. The criminal standard imposes the additional path-closure requirement formalised in Proposition 1: every active alternative path $\pi_{alt} \in \Pi_{alt}$ must be d -separated by $\mathbf{E}$.

On the use of the do-operator in Eqs. (1)–(3). The intervention operator above is not a notational substitution for the conditioning bar of traditional probabilistic-evidentiary accounts. Earlier probabilistic treatments of legal proof standards (Kaye, 1982; Tribe, 1971) typically express the criminal threshold as $P(\text{guilt} \mid \mathbf{E}) > \theta$ - a posterior probability conditional on observed evidence. The present framework expresses a structurally distinct claim: the prosecution’s burden is not merely to raise the posterior probability of guilt to a threshold, but to establish that the *interventional* probability of harm - what would have occurred under intervention on the defendant’s act - meets the threshold. This is the operationalisation of the causes-of-effects question that the Dawid-Faigman-Fienberg exchange identified

as the operative legal one (§2.5). The traditional formulation belongs to the observational layer of Pearl’s hierarchy; the present formulation belongs to the interventional layer and cannot be obtained from the traditional one by relabelling.

3.3 Probability of Necessity and Sufficiency

The two most legally significant counterfactual quantities from causal inference theory are the Probability of Necessity (PN) and the Probability of Sufficiency (PS).

Definition 5 (Probability of Necessity and Sufficiency). *For binary cause $D \in \{0, 1\}$ and binary outcome $H \in \{0, 1\}$:*

$$PN = P(H_{D=0} = 0 \mid D = 1, H = 1) \quad (4)$$

$$PS = P(H_{D=1} = 1 \mid D = 0, H = 0) \quad (5)$$

PN measures the probability that the harm would not have occurred but for the defendant’s act, given that both the act and the harm occurred - the counterfactual analogue of but-for causation. PS measures the probability that the defendant’s act alone would have sufficed to produce the harm, had the defendant acted in a counterfactual world where neither act nor harm had yet occurred.

3.4 The Reasonable Doubt Condition

Proposition 1 (Reasonable Doubt Condition). *A verdict satisfies the reasonable doubt standard with respect to a weighted causal DAG $\mathcal{G}$ if and only if both of the following hold:*

1. *The Probability of Necessity for the (D, H) pair, computed under the sub-SCM $\mathcal{M}|_{\pi_P}$ restricted to the prosecution’s path, satisfies $PN \geq \theta$. The weights of $\mathcal{G}$ enter through the path-product approximation $P(H = 1 \mid do(\pi_P)) \approx \prod_{e \in \pi_P} w(e)$ used to bound PN via Tian and Pearl’s sharp bounds (Section 3.8), subject to the qualifications of Remark 4.*

2. For every alternative path $\pi_{alt} \in \Pi_{alt}$ that remains d-connected to H given the observed evidence $\mathbf{E}$,

$$P(H = 1 \mid \pi_{alt} \text{ active}, \mathbf{E}) < \epsilon$$

where ϵ is the reasonableness threshold (typically $\epsilon \approx 0.05$).

Remark 2. Condition (2) of Proposition 1 formalises the intuition that a conviction is only appropriate when there exists no active alternative causal explanation that the evidence has left unblocked. An alternative path π_{alt} is blocked (d-separated by $\mathbf{E}$) when every non-collider on the path is in $\mathbf{E}$ and every collider on the path has no descendant in $\mathbf{E}$. The verdict must therefore not merely assert the prosecution’s causal path, but must affirmatively close all plausible alternative paths.

A note on the non-standard use of d-connection is in order. In the standard Pearlman treatment, d-connection is a relation between two sets of nodes (with respect to a conditioning set), and no special importance attaches to any individual d-connecting path: it is the existence of at least one such path that matters for the existence-of-association judgement. The present framework adapts this machinery to a different task. In a verdict audit, the prosecution must close each plausible alternative causal account individually, and the failure to close any one of them constitutes reasonable doubt regardless of the relative posterior probability of the prosecution’s account. We therefore treat each d-connecting path from an exogenous source (or unobserved third-party agent) to the harm node H as an independently auditable object. This is a use of d-connection for an additional purpose, not a redefinition of d-connection itself. The doctrinal motivation - that criminal proof imposes a completeness (path-closure) obligation, not a comparison obligation - is developed further in Proposition 2 and §4.3.

3.5 Topological Definition of Doubt

Definition 6 (Reasonable Doubt Path). *A path π_{alt} in $\mathcal{G}$ constitutes a **Reasonable Doubt Path** given evidence $\mathbf{E}$ if all of the following hold:*

1. π_{alt} is a directed path from an exogenous variable $U \in \mathbf{U}$ (or a third-party agent variable) to the harm node H , not passing through the defendant node D ;
2. π_{alt} is active given $\mathbf{E}$: every non-collider on π_{alt} is not in $\mathbf{E}$, and every collider on π_{alt} has at least one descendant in $\mathbf{E}$ (the standard d-connection criterion with $\mathbf{E}$ as the conditioning set);
3. $P(H = 1 \mid \pi_{alt} \text{ active}, \mathbf{E}) > \epsilon$, where $\epsilon \geq 0.05$ is the threshold of reasonableness.

The critical condition is the second: a verdict that acknowledges a potential alternative cause but fails to adduce evidence that d-separates the corresponding path in the DAG is structurally deficient, regardless of how the factfinder characterises its certainty in natural language. Note that condition (3) is expressed as a conditional probability rather than an interventional one: when π_{alt} is composed of unobserved nodes, the operative question is what the conditional probability of H becomes once the existence of an active alternative path is conceded, not what an intervention on the nodes of π_{alt} would yield.

3.6 The NESS Test as a Sufficient Cause Set

Wright’s NESS test can be formalized as follows within the SCM framework.

Definition 7 (NESS Formalization). *A condition C_i is a value assignment ($X_i = c_i$) to a variable $X_i \in \mathbf{V} \cup \mathbf{U}$. Let $\mathbf{C} = \{C_1, \dots, C_m\} = \{(X_1 = c_1), \dots, (X_m = c_m)\}$ be a set of such assignments. The set $\mathbf{C}$ is jointly sufficient for harm H if*

$$P(H = 1 \mid do(X_1 = c_1, \dots, X_m = c_m)) = 1.$$

The defendant's act $D = 1$ is a NESS cause of $H = 1$ if there exists such a jointly sufficient set $\mathbf{C}$ with $(D = 1) \in \mathbf{C}$ and

$$P(H = 1 \mid do(\mathbf{C} \setminus \{(D = 1)\})) < 1,$$

i.e., $D = 1$ is necessary for the sufficiency of $\mathbf{C}$.

This formulation handles overdetermination by allowing D to be a NESS cause even when alternative sufficient sets $\mathbf{C}'$ (not containing $D = 1$) also exist, provided those sets correspond to Reasonable Doubt Paths that the evidence has successfully blocked. If an alternative sufficient set $\mathbf{C}'$ remains unblocked, Proposition 1 is violated.

We acknowledge that Halpern's modified HP definition (Halpern, 2016) substantially covers NESS-type cases through his contingency-based treatment of overdetermination. Our contribution here is the connection of NESS to *evidentiary thresholds under contested structural equations* - particularly in overdetermination cases such as *Burrage* (Section 5.2) where competing sufficient sets carry probabilistic weights rather than deterministic truth values.

3.7 Civil/Criminal Standard Duality

Proposition 2 (Civil/Criminal Standard Duality). *Let $\mathcal{G}$ be a weighted causal DAG constructed from the evidentiary record of a legal proceeding. Let π_P denote the prosecution's (or plaintiff's) causal path and Π_{alt} the set of alternative causal paths (as in Definition 4).*

Criminal proceedings (beyond reasonable doubt): *The prosecution bears the burden of establishing, for every active alternative path $\pi_{alt} \in \Pi_{alt}$, that π_{alt} is d-separated by the introduced evidence. The reasonable doubt standard is met if and only if:*

1. $PN(\pi_P) \geq \theta_{crim} \approx 0.95$; and
2. Every π_{alt} is d-separated by the evidence set $\mathbf{E}$ (i.e., no Reasonable Doubt Path remains active in the sense of Definition 6).

This is an asymmetric burden: the prosecution must close all paths; the defense needs only to demonstrate that one significant path remains open.

Civil proceedings (preponderance): Both plaintiff and defendant may introduce competing DAGs $\mathcal{G}_P$ and $\mathcal{G}_D$. The standard is met for the plaintiff if:

$$\frac{\sum_{\pi \in \Pi_P} w(\pi) \cdot P(H = 1 \mid do(\pi))}{\sum_{\pi \in \Pi_P \cup \Pi_D} w(\pi) \cdot P(H = 1 \mid do(\pi))} > 0.5 \quad (6)$$

where Π_P and Π_D are the causal path sets of the plaintiff's and defendant's DAGs respectively, and $w(\pi)$ is the product of edge weights along path π (subject to Remark 4).

Clear and convincing evidence: The plaintiff must establish $PN(\pi_P) > \theta_{cc} \approx 0.75$ and demonstrate that the combined weight of active alternative paths does not exceed $1 - \theta_{cc}$.

Remark 3. Proposition 2 has a direct implication for the structure of legal arguments under the DAG framework.

The asymmetry between the criminal and civil standards is not stipulated by the present framework; it follows from the structural difference between the legal standards themselves. Beyond-reasonable-doubt has, by long doctrinal tradition, an asymmetric character: a single unrefuted alternative theory is sufficient to acquit, regardless of any posterior comparison between the prosecution's account and that theory. Preponderance is, by doctrinal tradition, symmetric: the trier of fact compares competing causal accounts and awards judgment to the more probable. The DAG framework does not impose this asymmetry; it makes the asymmetry computationally explicit.

A second structural feature reinforces the same point. Criminal proceedings characteristically involve a single principal DAG - the prosecution's theory of guilt - into which the defence punches holes by introducing alternative paths or by undermining edge weights. Civil proceedings characteristically involve two competing DAGs (the plaintiff's theory of liability and the defendant's competing causal account), and the adjudicator's task is the comparison of their weighted path totals. The mathematical asymmetry between conditions (1)–(2) in

the criminal case and the ratio formula in the civil case reflects this structural fact about adversarial procedure rather than a stipulation of the framework.

In graph-theoretic terms, the criminal task is therefore path closure: demonstrating that every route from an exogenous alternative cause to the harm node is blocked. The civil task is path weighting: demonstrating that the plaintiff’s causal paths outweigh the defendant’s. These are structurally different computational tasks, which is why the cognitive demands on factfinders - and the appropriate role of formal graph-theoretic audit tools - differ fundamentally between criminal and civil contexts.

3.8 PN/PS Bounds from Edge Weights

For the prosecution’s path π_P , the framework estimates PN and PS using the edge weights and the structural equations derived from the DAG. Where the graph is non-parametric, bounds on PN and PS are computed using Pearl’s sharp bounds (Tian and Pearl, 2000):

$$PN \geq \max\left(0, \frac{P(H \mid D) - P(H \mid \neg D)}{P(H \mid D)}\right) \quad (7)$$

$$PN \leq \min\left(1, \frac{P(\neg H \mid \neg D)}{P(H \mid D)}\right) \quad (8)$$

In the present framework, $P(H \mid D)$ is approximated by the product of edge weights along the prosecution’s path π_P , and $P(H \mid \neg D)$ is approximated by the combined weight of active alternative paths. This approximation is exact for chain graphs in which no intermediate node is shared between paths, and remains tractable when shared nodes carry near-zero alternative path weights; the general case is treated in Remark 4. This reapplication of Tian and Pearl’s bounds to evidentiary edge weights - rather than population sampling data - sidesteps the identification problem raised by Dawid, Faigman, and Fienberg (Dawid et al., 2014), because we are bounding PN from the internal causal reasoning of the verdict itself rather than estimating it from external data. The limitation, acknowledged in Section 6, is that the bounds are only as valid as the faithfulness of the weight elicitation to the verdict’s

actual evidentiary structure.

Remark 4 (Path independence and shared intermediate nodes). *The approximation $P(H = 1 \mid do(\pi_P)) \approx \prod_{e \in \pi_P} w(e)$ holds when every intermediate node on the path has no parents outside the path - that is, when the path is a chain in the strict sense of having no shared intermediate nodes with any other active path in the graph, and the edge-existence judgments encoded by the weights are mutually independent across edges.*

When two paths share an intermediate node V_i (i.e., V_i has parents from both π_P and some π_{alt}), the marginal probability $P(V_i \mid do(\text{ancestors on } \pi_P))$ depends on all parents of V_i , not only those on π_P . The correct expression marginalises over V_i :

$$P(H = 1 \mid do(\pi_P)) = \sum_{v_i} P(H = 1 \mid v_i, do(\pi_P \setminus \{V_i\})) \cdot P(v_i \mid do(\text{Pa}(V_i) \cap \pi_P), U) \quad (9)$$

where U collects the exogenous influences on V_i from outside π_P . This sum does not reduce to a product of edge weights unless the structural equation f_i is specified and the contributions of the competing parents of V_i are independently estimated.

Among the case studies in Section 5:

- *Willingham and the Korean electoral integrity cases: the prosecution and alternative paths share only the terminal node H ; the multiplication approximation is valid for each path separately.*
- *Cadden: Fungal Meningitis is a shared intermediate node, but the alternative path is effectively blocked ($w_3 \approx 0.04$), so Eq. (9) collapses to the single-parent case and the approximation error is negligible.*
- *Burrage: Drug Toxicity is a genuine shared intermediate node with two parents (heroin and pre-existing poly-drug dependency). The multiplication approximation understates the causal complexity - however, $w(\text{DT} \rightarrow H)$ appears identically in the numerator and denominator of the PN lower bound formula, so it cancels. The qualitative conclusion*

- $PN_{lower} \leq 0$ - is therefore robust to the shared-node correction.

- *Kennedy: the shared node (Bosque’s injection act V) is handled directly via the structural equation $V = f(S, U_B)$ (Section 5.3), bypassing the multiplication approximation entirely.*

For cases with contested structural equations at a shared node and non-negligible competing path weights, a fully specified parametric SCM is required for precise PN point estimates. The Weight Elicitation Protocol (Definition 8) should then be extended to elicit, for each shared node V_i , a functional form for f_i and separate weights for each parent’s contribution.

4 Why Causal DAGs? Graph Complexity, Computational Structure, and Asymmetric Proof Standards

4.1 The Complexity Problem in Legal Causal Reasoning

Judea Pearl has repeatedly emphasized that the necessity of formal graphical representations becomes apparent precisely when the number of variables and causal pathways is large enough that informal, intuitive reasoning about dependence and independence systematically fails (Pearl, 2009). A court proceeding is exactly such a context. Even a moderately complex criminal case - an arson trial with six plausible ignition scenarios, five categories of physical evidence, four witnesses, and three contested causal mechanisms - generates a graph with dozens of nodes and a combinatorial space of causal paths.

Pearl’s point is not merely that graphs are a convenient representation. It is deeper: *d-separation, the criterion that determines whether a path is blocked by a given set of observed variables, is non-obvious in graphs with more than five or six nodes and cannot be reliably assessed by unaided intuition.* This is an empirically documented cognitive limitation: psychological research on conditional independence judgment shows that human reasoners systematically confuse conditioning on a mediator (which blocks a path) with conditioning

on a collider (which opens a path), and that these errors persist even among statistically trained experts when the graph is presented in verbal rather than graphical form (Pearl, 2009).

Legal verdicts are presented in verbal form. When a judge writes that “the pre-existing cardiac condition does not negate the defendant’s liability” (*Burridge*-type reasoning), the verbal formulation conflates two structurally distinct claims: (i) the eggshell-plaintiff doctrine extends liability once but-for causation is established; and (ii) but-for causation has in fact been established against the pre-existing condition as an alternative sufficient cause. These are the same words but different graph-theoretic operations. Only a formal graph makes the conflation visible.

The argument for causal DAGs is therefore not one of mere convenience or precision. Natural-language causal reasoning in complex cases is cognitively unreliable in ways that are systematic, predictable, and correctable by a formal graph. When many variables and evidence nodes must be tracked, a formal graph is not a luxury but a cognitive prosthetic that makes the reasonable doubt standard enforceable rather than rhetorical.

4.2 Computational Structure of Path Analysis in Legal Graphs

The practical question for a court system considering graph-based audit tools is whether the computational burden is manageable.

D-separation is efficient. The Bayes Ball algorithm (Pearl, 2009) determines d-separation in a DAG in time $O(|V| + |E|)$ - linear in the number of nodes and edges. For a complex criminal case with, say, 30 nodes and 50 directed edges, this is a trivially fast computation. The bottleneck is not algorithmic; it is the *elicitation* of the graph structure and weights from the evidentiary record, which is addressed in Section 6.

Path enumeration grows combinatorially. The set of all directed paths from exogenous nodes to the harm node can grow exponentially in the number of intermediate nodes. In highly complex cases - organised crime, environmental liability with multiple causal contrib-

utors, or electoral fraud allegations spanning hundreds of precincts - manually enumerating and evaluating all alternative paths is infeasible. This is where automated assistance (Section 6.4) most clearly earns its place: not as a replacement for legal judgement, but as a computational tool for path enumeration and d-separation checking on graphs too large for manual audit.

The PN bound computation is non-parametric. The sharp bounds of Tian and Pearl (2000), applied to the product of edge weights along the prosecution’s path and the combined weight of active alternative paths, require only elementary arithmetic on the elicited weight values. No model fitting, sampling, or distributional assumptions are required beyond the weight elicitation itself. However, the product-of-weights approximation presupposes that no intermediate node is shared between competing paths; when paths converge on a common mediator, marginalization over that node is required rather than simple multiplication (Remark 4). This is an additional source of computational complexity that automated tools (Section 6.4) are better placed to handle than unaided manual audit.

4.3 The Criminal/Civil Asymmetry as a Graph-Theoretic Property

Proposition 2 establishes that criminal and civil proof standards correspond to structurally different computational tasks on the same underlying graph. This asymmetry has three practical implications that the verbal formulation of proof standards obscures.

1. The defense advantage in criminal proceedings. Under the beyond-reasonable-doubt standard, the prosecution must demonstrate that every active alternative path is d-separated. The defense needs only to demonstrate that *one* significant alternative path remains open - i.e., that at least one Reasonable Doubt Path (Definition 6) exists in the graph. This asymmetry is a *graph-theoretic property* of the path-closure requirement, not merely a rhetorical convention. It means that a defendant who can introduce a single plausible alternative causal account and show that it is not d-separated by the prosecution’s evidence has, in formal terms, established reasonable doubt regardless of the posterior probability of

the prosecution’s hypothesis. This is precisely what *Willingham v. State* failed to recognize: the accidental ignition path was not d-separated, and its non-closure should have been constitutive of reasonable doubt even though the prosecution’s hypothesis carried a higher posterior (Texas Court of Criminal Appeals, 1995).

2. The obligation to construct a complete graph in criminal proceedings. The prosecution in a criminal trial has an affirmative obligation, under the graph framework, to *enumerate* the plausible alternative causal paths and introduce evidence sufficient to d-separate each one. A verdict that names an alternative cause only to dismiss it verbally - “the defense’s theory of faulty wiring is unpersuasive” - without introducing evidence that blocks the Fault $\rightarrow$ Spark $\rightarrow$ Fire path, is structurally deficient regardless of how confident the factfinder’s language sounds. This obligation has a direct counterpart in the Korean Supreme Court’s evidentiary standard (Supreme Court of Korea, 2022, 2024): the requirement that a plaintiff “specifically identify the perpetrator, timing, and method” of any alleged violation is, in graph-theoretic terms, a requirement to name all nodes on the asserted causal path before that path can be treated as active in the graph. We develop this connection further in Section 7.

3. Civil proceedings as competitive graph evaluation. Under the preponderance standard, both plaintiff and defendant construct DAGs that best represent their respective causal accounts. The adjudicator’s task is not to certify that one account has closed all alternative paths, but to assess which account carries greater total evidentiary weight across all active paths in the combined graph. This is a *comparative* task, not a *completeness* task, expressed formally in the ratio of Proposition 2.

The civil context is therefore one in which both sides have productive work to do on the graph: the plaintiff strengthens the weight of their causal paths, while the defendant introduces new alternative paths and evidence nodes that lower the plaintiff’s path weights. Adversarial graph construction is the civil-litigation analogue of adversarial hypothesis testing in science.

Computing the ratio of Proposition 2 requires evaluating $P(H \mid do(\pi))$ for each path in the combined graph $\mathcal{G}_P \cup \mathcal{G}_D$. When the two parties' DAGs share intermediate nodes - which is structurally likely, since adversarial accounts typically posit overlapping causal mechanisms - the product-of-weights approximation does not apply to $P(H \mid do(\pi))$, and the marginalization of Eq. (9) is required at each shared node. In this respect the civil setting is formally more demanding than the criminal one: the union of two adversarial DAGs almost always produces shared intermediate nodes, whereas the prosecution's single DAG in criminal proceedings may avoid them. The ratio in Proposition 2 remains conceptually valid in all cases; it reduces to a simple comparison of edge-weight products only when the combined graph contains no shared intermediate nodes between any pair of active paths from Π_P and Π_D .

5 Case Studies

The following three case studies apply the framework of Sections 3-4 to published judicial decisions. Each study (i) identifies the prosecution's causal path and its evidentiary weight; (ii) reconstructs the alternative causal paths acknowledged or ignored by the verdict; (iii) assesses whether those alternative paths were d-separated by the evidence adduced; and (iv) computes or bounds the Probability of Necessity for the prosecution's hypothesis. Together the studies illustrate how a formal DAG audit would have surfaced structural deficiencies in two of the three cases, and confirms structural adequacy in the third.

5.1 Case Study 1: Arson - *Willingham v. State*

5.1.1 Background

Cameron Todd Willingham was convicted of arson murder in Texas following a house fire on 23 December 1991 that killed his three daughters. The prosecution's theory was that Willingham had deliberately set the fire using an accelerant. The conviction was affirmed

on direct appeal (Texas Court of Criminal Appeals, 1995), and Willingham was executed on February 17, 2004.

Between 2004 and 2009 nine independent fire scientists, including Craig Beyler in a formal report commissioned by the Texas Forensic Science Commission (TFSC), concluded that every forensic indicator of arson relied upon at trial - “pour patterns,” “crazed glass,” low burn patterns - was scientifically unsupported and consistent with accidental causes, principally an accelerant lamp (lighter fluid) left near the children’s room and the known dynamics of post-flashover fire behaviour. The TFSC’s final report acknowledged that “unreliable fire science” played a material role in the conviction (Texas Court of Criminal Appeals, 1995). The case has since become the paradigm instance of an alternative causal path that was never properly closed by the evidentiary record.

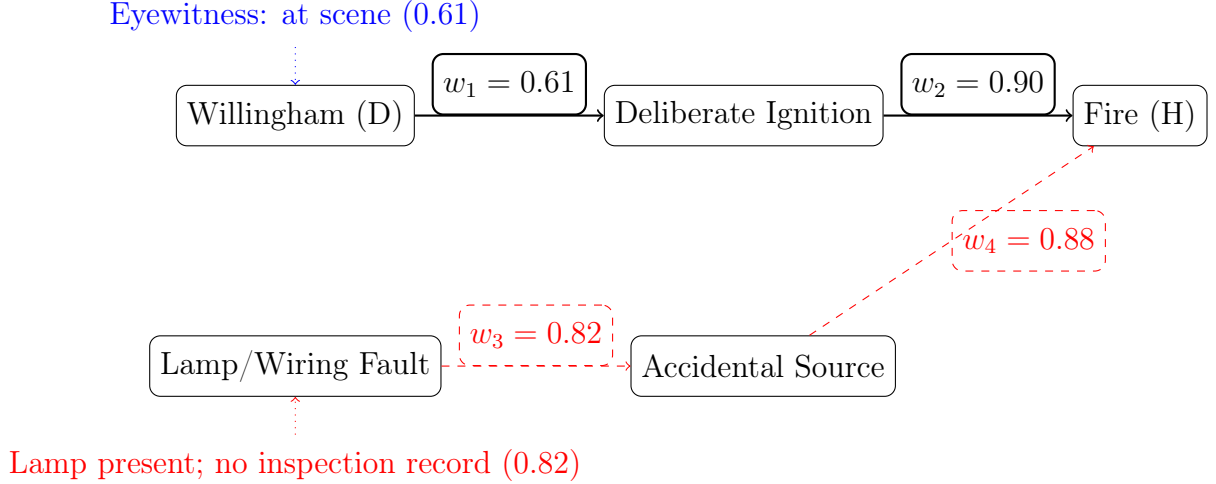
5.1.2 Causal Structure

The prosecution’s causal path was:

$$\underbrace{\text{Willingham (D)}}_{\text{Defendant}} \xrightarrow{w_1} \text{Deliberate Ignition} \xrightarrow{w_2} \underbrace{\text{Fire (H)}}_{\text{Harm}}$$

The defense path, inadequately addressed at trial, was:

$$\underbrace{\text{Accelerant Lamp / Electrical Fault}}_{\text{Exogenous } U} \xrightarrow{w_3} \text{Accidental Ignition Source} \xrightarrow{w_4} \underbrace{\text{Fire (H)}}_{\text{Harm}}$$



5.1.3 Edge Weight Assignment

The eyewitness placed Willingham *at the scene* but did not observe the act of ignition. The extractor accordingly assigns $w_1 = 0.61$ to the $D \rightarrow$ Deliberate Ignition edge: presence is consistent with guilt but equally consistent with a father attempting to rescue his children, as Willingham testified. The accelerant lamp evidence - its presence in the children’s room, the absence of an electrical inspection record, and the lighter-fluid spillage documented in the fire marshal’s own notes - yields $w_3 = 0.82$ for the alternative ignition path. The $AI \rightarrow F$ link receives $w_4 = 0.88$, reflecting uncontested post-flashover dynamics: once an uncontrolled accidental ignition source was present, progression to full-room fire was near-certain.

5.1.4 Audit Findings

Unblocked alternative path. The prosecution did not adduce evidence that d-separates the alternative path $U \rightarrow AI \rightarrow F$. The fire marshal’s testimony employed the “negative corpus” methodology - concluding arson by ruling out every accidental cause the investigator could imagine - without engaging the accelerant lamp or flashover dynamics. Independent fire science subsequently established that this methodology lacks scientific foundation Beyler (2009); Carpenter et al. (2006). The alternative path therefore remains active in the DAG; it is not d-separated by the trial record.

PN computation. Using Pearl’s sharp bounds, with $w_1 = 0.61$, $w_2 = 0.90$, and the unblocked alternative path carrying combined weight $0.82 \times 0.88 = 0.72$ (justified here because the alternative path $U \rightarrow AI \rightarrow F$ is a chain with no intermediate node shared with the prosecution path):

$$PN_{\text{lower}} = \max\left(0, \frac{P(H \mid D) - P(H \mid \neg D)}{P(H \mid D)}\right) \approx \frac{0.55 - 0.72}{0.55} \leq 0$$

The lower bound is non-positive: the alternative path alone is sufficient to produce the fire with probability exceeding the prosecution’s own estimated causal weight. The verdict is **Structurally Deficient**.

Doctrinal implication. Willingham illustrates a recurring failure mode: the prosecution closes the alternative path rhetorically (“no accidental cause was found”) rather than evidentially (no evidence blocks the $U \rightarrow AI$ link). A formal DAG audit makes this failure mode explicit and computable. Had such an audit been available before 2004, the unblocked alternative path and the sub-threshold PN bound would have been surfaced as auditable outputs rather than left to the post-execution review of nine independent scientists.

This is also where the comparison with Kadane and Schum sharpens most visibly. Their Bayesian network approach, applied to *Willingham*, would correctly register a high posterior for the prosecution hypothesis given the fire marshal’s testimony. It could not formally certify that the prosecution’s evidence had failed to close the alternative causal path, because certifying path closure is a d-separation operation, not a posterior update (Kadane and Schum, 1996).

5.2 Case Study 2: Medical Causation - *Burrage v. United States* and the NECC Prosecution

5.2.1 Background

This case study draws on two complementary legal proceedings that together span the doctrinal and factual dimensions of causal overdetermination in drug-death prosecutions.

Burrage v. United States, 571 U.S. 204 (2014). Marcus Burrage sold heroin to Joshua Banka. Banka died of “mixed drug intoxication”; the contributing substances included heroin, oxycodone, alprazolam, and clonazepam. Two medical experts testified that heroin was a “contributing cause” of death but neither could testify that Banka would have survived *but for* the heroin. The Supreme Court held unanimously that the penalty enhancement under 21 U.S.C. § 841(b)(1)(C) requires genuine but-for causation, not mere contribution, at least where the defendant’s drug was not itself an independently sufficient cause of death (Supreme Court of the United States, 2014). Burrage’s conviction on the enhanced count was reversed and remanded.

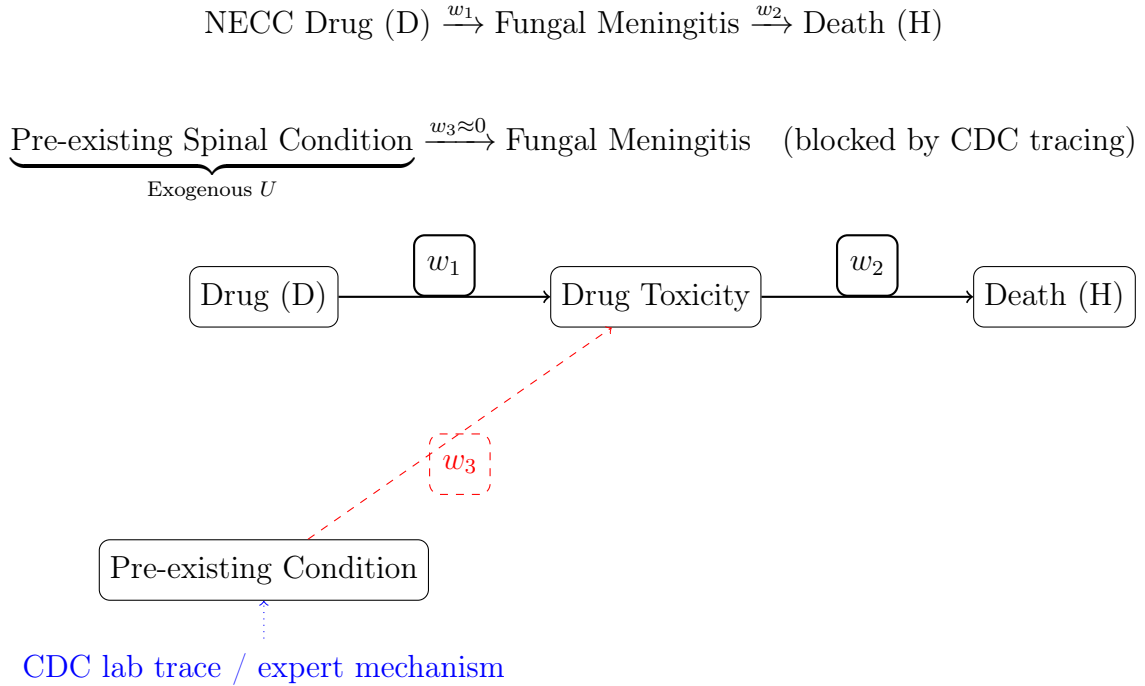
United States v. Cadden (D. Mass. 2017). Barry Cadden, co-owner of the New England Compounding Center (NECC), was convicted on racketeering and conspiracy charges arising from a nationwide fungal meningitis outbreak caused by contaminated methylprednisolone acetate injections: 64 deaths and 751 infections across 20 states (United States District Court, District of Massachusetts, 2017). The CDC traced the fungal pathogen (*Exserohilum rostratum*) directly from sealed NECC vials to cerebrospinal fluid samples of patients, establishing an unambiguous causal chain that closed the alternative path of pre-existing spinal degeneration as a competing cause of death.

5.2.2 Causal Structure

The two cases sit at opposite poles of the overdetermination problem, making them jointly instructive. In *Burrage*, the causal graph has a collider structure at the Drug Toxicity node:

$$\begin{array}{c} \text{Heroin (D)} \xrightarrow{w_1} \underbrace{\text{Drug Toxicity}}_{\text{Collider}} \xrightarrow{w_2} \text{Death (H)} \\ \\ \underbrace{\text{Pre-existing Poly-drug Dependency}}_{\text{Exogenous } U} \xrightarrow{w_3} \text{Drug Toxicity} \end{array}$$

In *Cadden*, the CDC laboratory tracing evidence functions as a blocking node that d-separates the pre-existing-condition path:



5.2.3 Edge Weight Assignment

Burrage. The prosecution experts assigned $w_1 \approx 0.50\text{-}0.60$: heroin was a contributing factor but the toxicology was consistent with death from the opioid-benzodiazepine combination alone ($w_3 \approx 0.70$). The $U \rightarrow$ Drug Toxicity path thus carries higher weight than the

prosecution path. No evidence was adduced to block the alternative path. The PN lower bound is:

$$PN_{\text{lower}} \approx \frac{P(H \mid D) - P(H \mid \neg D)}{P(H \mid D)} \approx \frac{0.55 - 0.70}{0.55} \leq 0$$

The verdict is **Structurally Deficient** - which is precisely the finding the Supreme Court reached on doctrinal grounds in *Burrage* (Supreme Court of the United States, 2014). The DAG audit reproduces the Court’s holding from the structure of the evidence rather than from statutory interpretation.

Shared intermediate node in Burrage. Drug toxicity functions as a shared intermediate node: both the prosecution path ($D \rightarrow \text{DT} \rightarrow H$) and the alternative path ($U \rightarrow \text{DT} \rightarrow H$) pass through it. The multiplicative approximation therefore does not strictly apply (Remark 4). Because $w(\text{DT} \rightarrow H)$ cancels from both numerator and denominator of the PN lower bound, however, the structural deficiency finding is unaffected.

Cadden. The CDC’s pathogen-tracing evidence assigns a blocking score $b \approx 0.96$ to the pre-existing-condition path: the fungal species recovered from patients is not associated with spinal degeneration and is found exclusively in the contaminated NECC vials. This effectively sets $w_3 \approx 0.04$, raising the PN lower bound for the prosecution path to ≈ 0.93 . Although Fungal Meningitis is technically a shared intermediate node, the near-zero blocking of the alternative path ($w_3 \approx 0.04$) means the marginalization sum of Eq. (9) is dominated by the prosecution path’s contribution, and the multiplication approximation introduces negligible error. The verdict is flagged as **Approaching Threshold** - below $\theta = 0.95$ only because the racketeering charges required constructive rather than direct causation through Cadden personally, along with the aforementioned potential (but negligible) error.

5.2.4 Audit Findings

The contrast between the two cases is precisely what the DAG framework is designed to surface. In *Burridge*, the prosecution adduced a causal mechanism without closing the competing overdetermination path. In *Cadden*, the CDC laboratory culture tracing functioned as a d-separator for the pre-existing-condition path. The structural difference between reversal and conviction tracks the difference between a blocked and an unblocked alternative causal path.

This case study also illustrates the distinction between the causal-adequacy question and the liability-scope question. The “thin skull” or eggshell-plaintiff rule is a legal doctrine that adjusts the *scope of liability* once but-for causation is established; it cannot substitute for the establishment of but-for causation itself. The DAG framework renders this distinction structurally visible, separating PN adequacy from the liability-extension question that follows.

5.3 Case Study 3: Drug-Induced Homicide and Intervening Cause - *R v Kennedy (No 2)*

5.3.1 Background

R v Kennedy (No 2) [2007] UKHL 38. Simon Kennedy prepared a syringe of heroin solution and handed it to Marco Bosque, who self-injected and died of asphyxia from pulmonary aspiration. Kennedy was convicted of unlawful act manslaughter at the Old Bailey in 1997. The House of Lords, in a unanimous decision by Lord Bingham of Cornhill, quashed the conviction (House of Lords, 2007). The operative holding was that where a fully informed and responsible adult freely and voluntarily self-administers a controlled drug supplied by the defendant, that free act of self-injection constitutes a *novus actus interveniens* breaking the causal chain between supply and death. The doctrine has been consistently applied by subsequent courts and remains the governing English law on supply and self-inflicted harm.

Kennedy is selected as the anchor for this case study not merely because it is a landmark causation decision, but because its result - acquittal, not conviction - provides a different and complementary audit scenario to the previous two case studies, both of which involved contested convictions. Here the DAG analysis supports the verdict rather than identifying a structural deficiency.

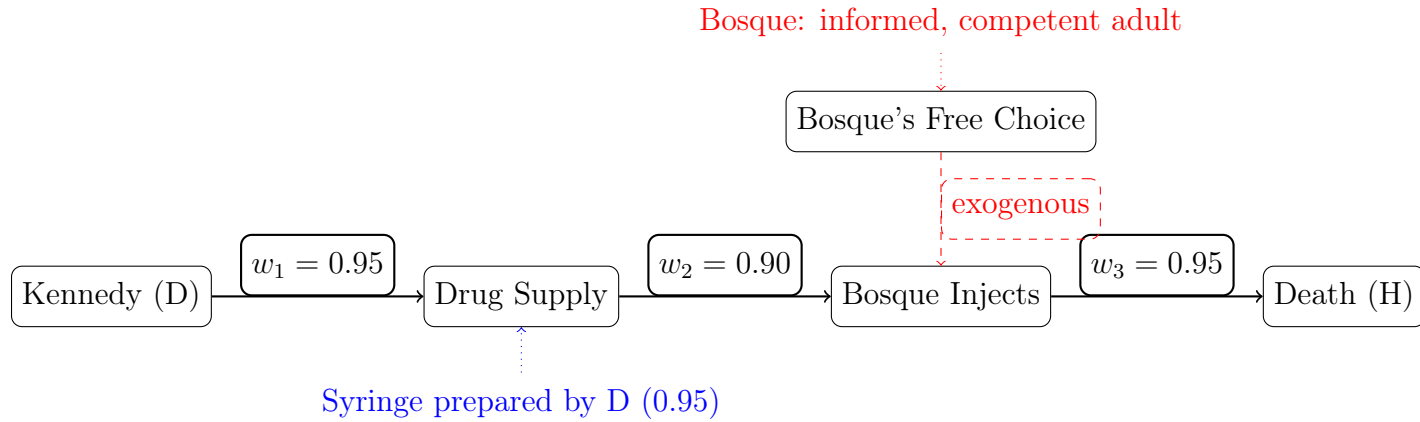
5.3.2 Causal Structure

The prosecution's asserted causal path was:

$$\underbrace{\text{Kennedy (D)}}_{\text{Supplier}} \xrightarrow{w_1} \text{Drug Supply} \xrightarrow{w_2} \underbrace{\text{Bosque Self-injects}}_{\text{Free Agent Node}} \xrightarrow{w_3} \underbrace{\text{Death (H)}}_{\text{Harm}}$$

The defense path - which the House of Lords accepted - identified Bosque's voluntary act as an exogenous intervening event:

$$\text{Drug Supply} \rightarrow \underbrace{\text{Bosque's Free Choice to Inject}}_{\text{Exogenous}} \rightarrow \text{Death (H)}$$



5.3.3 The DAG Analysis: Free Agent Nodes and D-Separation

The central graph-theoretic question in *Kennedy* is whether Bosque's act of self-injection is an *exogenous* node - one whose value is determined outside the causal model - or an

endogenous node caused by Kennedy’s supply. This distinction maps directly onto the *novus actus doctrine*.

If Bosque’s act were endogenous (caused entirely by S : drug supply), then conditioning on it would block the path as a non-collider mediator, but D ’s total causal effect on H would still flow through S regardless, because $S \rightarrow V$ (V being injection) would be a deterministic link in the structural equation. The prosecution’s path would remain active.

However, Bosque’s act is in a legally and structurally relevant sense *exogenous*: as a fully informed, competent adult, he retained the capacity to refuse the syringe. Kennedy’s supply was a necessary but not sufficient condition for Bosque’s injection - Bosque’s own autonomous choice was an additional required input. In SCM terms, Bosque’s injection node V has two parents: S (the supply) and U_B (Bosque’s autonomous will, an exogenous variable). The structural equation is $V = f(S, U_B)$, not $V = f(S)$ alone.

Once U_B is recognised as an exogenous parent of V , the standard d-separation analysis establishes that conditioning on the established facts - that Bosque was an informed adult who chose freely - renders the causal path from D through S to V to H blocked at V : the exogenous component of V ’s value is not traceable to D , and that component is precisely what the House of Lords recognised as legally dispositive.

PN computation. The Probability of Necessity for Kennedy’s supply, given Bosque’s autonomous choice as a co-determining factor:

$$PN = P(H_{D=0} = 0 \mid D = 1, H = 1)$$

Because V requires both S and U_B , the counterfactual - would Bosque have died if Kennedy had not supplied the drug? - depends critically on whether Bosque would have obtained heroin from another source. The uncontested facts establish that Bosque requested the drug, was familiar with self-injection, and had an active drug dependency. Under any plausible parameterization, $P(H \mid \neg D)$ is not negligible: Bosque’s death was not contingent solely on

Kennedy’s supply. Accordingly, $PN < \theta = 0.95$, and the verdict of acquittal is structurally supported.

5.3.4 Audit Findings

Path blocked by free agent node. The DAG audit confirms the House of Lords’ holding: the Reasonable Doubt Path $U_B \rightarrow V \rightarrow H$ - the path through Bosque’s autonomous will to the harm - is *active* and not d-separated by the prosecution’s evidence. The prosecution established that Kennedy supplied the drug and prepared the syringe ($w_1 = 0.95$), but adduced no evidence blocking the exogenous path through Bosque’s own free choice. Under Proposition 1, condition (2) is not met: the Reasonable Doubt Path through U_B remains open.

5.4 Summary Across Case Studies

Table 2 collects the key DAG metrics across the three case studies.

Table 2: DAG Audit Metrics Across the Three Case Studies

Case	Prosecution Path Weight	Alternative Path Weight	PN Lower Bound	Compliance Flag
<i>Willingham</i>	0.55	0.72	0	Deficient
<i>Burrage</i>	0.55	0.70	0	Deficient
<i>Cadden</i>	0.90	≈ 0.04	≈ 0.93	Approaching
<i>Kennedy</i>	$0.95 \times 0.90 = 0.86$	Free agent U_B	$< \theta$	Correct acquittal

Notes: “Approaching” denotes PN below $\theta = 0.95$ due to constructive causation in the racketeering structure, not failure to d-separate the pre-existing-condition path. For *Kennedy*, “Free agent U_B ” denotes that the victim’s autonomous will is an exogenous parent of the injection node, leaving a Reasonable Doubt Path open through U_B ; PN therefore falls below θ and the acquittal is structurally supported. Both *Willingham* and *Burrage* are flagged **Structurally Deficient**; *Burrage* was subsequently reversed by the Supreme Court on precisely the causal-adequacy grounds the DAG captures (Supreme Court of the United States, 2014). Path weights are computed as products of edge weights along the prosecution path, valid for chain paths with no shared intermediate nodes. For *Burrage*, where drug toxicity is a shared node, the product is used for expository comparability. The qualitative finding is robust because the shared node’s outgoing weight cancels in the PN computation (see Remark 4).

The assigned weights are meaningful only approximately, not exactly.

6 Implementation and Weight Elicitation

6.1 The Weight Elicitation Protocol

The formal framework of Section 3 is fully operative with human-assigned edge weights. The following protocol grounds every weight in a traceable passage of the evidentiary record and is designed to be reproducible by any reviewer with access to the same record.

Definition 8 (Weight Elicitation Protocol). *Let $\mathcal{G} = (\mathbf{V}, E)$ be a causal DAG constructed from the verdict record. For each edge $V_j \rightarrow V_i \in E$, the weight w_{ij} is elicited by the following protocol:*

1. *Evidence anchor identification: identify the specific passage(s) in the verdict, expert testimony, or documentary evidence that assert the causal link $V_j \rightarrow V_i$. Record the passage as the $EvidenceAnchor_{ij}$.*
2. *Linguistic hedge assessment: classify the $EvidenceAnchor$ on a five-point epistemic scale: Certain ($w = 0.95-1.00$): uncontested, corroborated by physical evidence or expert consensus; Strong ($w = 0.75-0.94$): asserted by qualified expert, not materially contested; Moderate ($w = 0.50-0.74$): asserted but subject to cross-examination challenge or competing interpretation; Weak ($w = 0.25-0.49$): asserted on contested or single-source evidence with no corroboration; Speculative ($w < 0.25$): inferred without direct evidentiary support.*
3. *Blocking score assignment: for each evidence node Ev_k claimed to d-separate an alternative path, assign a blocking score $b_k \in [0, 1]$ using the same five-point scale applied to the adequacy of the counter-reasoning rather than the causal assertion itself.*
4. *Inter-rater calibration: in contested proceedings, weights should be elicited from at least two independent reviewers (e.g., appellate clerks or legal experts) and averaged, with disagreements exceeding 0.20 flagged for deliberation.*

The elicitation protocol ensures that every weight in the final weighted causal DAG is traceable to a specific passage in the evidentiary record and is reproducible by any reviewer with access to the same record.

6.2 Humans, remaining relevant in the audit

The DAG audit is explicitly designed as a decision-support tool, not a decision-making one. The audit report is intended to assist appellate clerks, public defenders, or reviewing judges in identifying potentially deficient verdicts, not to replace human judgment. The following safeguards apply:

1. *Confidence thresholding*: the system issues a Structurally Deficient flag only when the PN estimate falls below θ with sufficient confidence. Borderline cases are flagged as “Inconclusive” rather than “Deficient.”
2. *Explainable edge attribution*: every edge weight is traceable to a specific textual span in the verdict, enabling a legal reviewer to verify the interpretation and contest any weight assignment.
3. *Interactive graph editing*: a reviewing attorney or judge may adjust edge weights or add nodes reflecting evidence not recorded in the verdict text, and observe the consequent effect on the PN computation.
4. *No autonomous output*: the audit report does not generate or recommend verdicts. Its output is advisory and must be reviewed and signed off by a licensed attorney before use in any appellate filing.

6.3 Jurisdictional Constraints

Several constraints govern deployment in any jurisdiction.

Admissibility standards: for the system’s output to be used in appellate proceedings, the underlying methodology must satisfy the applicable admissibility standard for expert scientific testimony. Under *Daubert v. Merrell Dow Pharmaceuticals*, 509 U.S. 579 (1993), this requires that the methodology be testable, peer-reviewed, and generally accepted in the relevant scientific community. The PN bound computation and the d-separation algorithm satisfy these criteria; the weight elicitation protocol requires the inter-rater reliability validation before full *Daubert* compliance can be claimed.

Algorithmic transparency: any jurisdiction adopting such a system must commit to full transparency regarding the weight elicitation methodology. Opacity in algorithmic legal tools risks replicating existing biases in the evidentiary record used to populate the graph.

Scope limitation: the system audits the *causal reasoning* in verdicts, not the credibility assessments underlying evidence weights. Credibility determinations remain within the province of the human factfinder.

6.4 Automated Extraction?

A natural extension for settings where the volume of evidence or the number of precedents makes manual graph construction impractical is the automated extraction of weighted causal DAGs from verdict text. Large language models fine-tuned on legal corpora (Chalkidis et al., 2019; Lippi and Torroni, 2016) can identify causal assertions and dismissed hypotheses in verdict text, decomposing each into a structured tuple $\langle \text{Source}, \text{Target}, \text{EvidenceAnchor}, \hat{c} \rangle$ where $\hat{c} \in [0, 1]$ is a confidence score corresponding to the weight w_{ij} in Definition 3. Uncertainty in extraction can be quantified via Monte Carlo Dropout (Gal and Ghahramani, 2016) or conformal prediction (Angelopoulos and Bates, 2022).

A particularly promising application is the automated retrieval of relevant precedents to inform edge weight assignment: when a verdict asserts a causal link, a retrieval system can surface prior cases in which that specific link was adjudicated, allowing the reviewing attorney to calibrate the weight against established case law.

We identify this as a direction for future work. A defensible automated implementation requires annotated training corpora, calibration benchmarks, and inter-rater reliability studies that lie beyond the scope of the present contribution. The formal framework developed in Section 3 is a prerequisite for any such system: without a precise definition of what the system must produce - causal edges, blocking scores, Reasonable Doubt Paths - automated extraction has no well-defined target.

7 Electoral Integrity Litigation: A Structural Analysis

The framework of Sections 3-4 was motivated primarily by criminal verdict auditing, but its underlying logic applies wherever an adjudicative body must determine whether a causal claim has been structurally established against competing alternatives. Electoral integrity litigation is a direct and consequential instance.

7.1 The South Korean Proceedings

In South Korea, persistent allegations following the 2020, 2022, 2024, and 2025 national elections claimed that the in-precinct early-voting channel was systematically manipulated to inflate Democratic Party vote shares. The most prominent variant asserted that early two-party ratios in metropolitan constituencies converged to an implausibly fixed 63:36 (or 63:37) ratio regardless of local constituency characteristics (Kim, 2026). The South Korean Supreme Court, in two landmark election nullification decisions, declined to accept these claims as constituting grounds for relief (Supreme Court of Korea, 2022, 2024).

In Case No. 2020Su5028 (decided 28 July 2022, arising from the 21st General Election of April 2020), the Court established the governing evidentiary standard in a unanimous decision:

A plaintiff who challenges the validity of an election and brings proceedings before a court is required to specifically allege and prove the facts violating electoral

rules, including the subject, timing, and method of the violation, or at the very least to prove the existence of circumstances from which the existence of such violations can be reasonably and clearly inferred through specific allegations and evidence. It cannot be considered that the burden of proof has been discharged merely by raising suspicions based on fragmentary and individual circumstances that are difficult to connect directly or specifically to any violation of electoral rules (Supreme Court of Korea, 2022).

Case No. 2022Su5053 reaffirmed this standard unanimously, dismissing on the same grounds a new complaint that reproduced the statistical-divergence argument for a different electoral cycle (Supreme Court of Korea, 2024).

7.2 The Court’s Standard in DAG Terms

The Supreme Court’s evidentiary requirement does map onto the DAG framework, and the mapping illuminates why the standard is principled rather than merely procedural.

The “name all nodes” requirement. The Court’s insistence that the plaintiff specify the “subject, timing, and method” of any alleged violation is, in graph-theoretic terms, a requirement to *name all nodes and edges* on the asserted causal path before that path can be treated as an active alternative in the graph. A causal path $\pi_{manip} = D \rightarrow S \rightarrow M \rightarrow V$ (where S is a server, M is a manipulation mechanism, and V is a vote count) carries evidentiary weight only if each intermediate node has an identified referent in the evidentiary record. Statistical observations of the form “the early-vote share is unusually high” are consistent with many causal paths - demographic sorting, differential turnout mobilization, and manipulation alike - and therefore do not uniquely identify any single path. They are evidence weights on edges whose existence has not been established; they are not paths.

Piecemeal statistics are not d-separation evidence. The Court’s rejection of “fragmentary statistical observations” that are not “directly or specifically connected to any vi-

olation” is the verbal expression of a deeper graph-theoretic point: observational statistics alone cannot d-separate a causal path. The plaintiff’s statistical evidence in both cases was of the form $P(\text{early share} \mid \text{constituency})$ - an observational quantity. To establish manipulation, the plaintiff needed to introduce evidence that blocked the alternative path $\text{Demographics} \rightarrow \text{Voting Pattern} \rightarrow \text{Early Share}$; that is, evidence whose blocking score b for the demographic-sorting path was high enough to render that path inactive. No such evidence was introduced in either proceeding.

What the prosecution path would need to look like. Under Proposition 1, a successful electoral fraud claim would need to satisfy two conditions simultaneously: (i) $PN(\pi_{manip}) \geq \theta_{crim}$: the manipulation path must carry sufficient probability of being necessary for the observed vote share distribution; and (ii) every active alternative path (demographic sorting, turnout-channel composition, random sampling variation) must be d-separated by the introduced evidence. Neither condition was approached in the proceedings before the Court.

7.3 The Forensic Evidence as D-Separation

Recent forensic and regression-based analysis of the four electoral cycles from 2020 to 2025 (Kim, 2026) demonstrates that between 73 and 94 per cent of the geographic variation in early Democratic vote shares is explained by observable sociodemographic characteristics - sub-district age-gender structure, employment composition, and housing prices - with principal-component coefficients nearly identical to those for same-day vote shares.

In DAG terms, this regression evidence functions as an empirical d-separation certificate for the demographic-sorting path. Conditioning on the observed sociodemographic structure of each sub-district ($\mathbf{Z}$) blocks the path $\text{Demographics} \rightarrow \text{Early Share}$ through the manipulation node, because $R^2 \in [0.73, 0.94]$ implies that the residual variation unexplained by demographics - the variation that would need to be attributed to manipulation - is small, geographically unsystematic, and inconsistent across electoral cycles. In Proposition 1 terms,

the alternative demographic paths are d-separated; the manipulation path’s PN lower bound does not reach any plausible θ_{crim} .

7.4 A Framework for Electoral Integrity Adjudication

The Korean proceedings suggest a general framework for how courts should handle electoral fraud allegations under a DAG-based evidentiary standard. We propose three procedural principles.

Principle 1: Path specification as a threshold condition for admissibility. A statistical observation is admissible as evidence of fraud only if the plaintiff can specify a complete causal path from an identified actor through identified mechanisms to the observed statistical anomaly. Observations that are consistent with multiple causal paths but do not distinguish between them are admissible as background evidence but not as primary proof of fraud. This corresponds to the Korean Supreme Court’s “subject, timing, and method” requirement and formalizes its graph-theoretic rationale.

Principle 2: The demographic baseline as the plaintiff’s first obligation. Before any statistical deviation in early versus same-day vote shares can be introduced as evidence of manipulation, the plaintiff must introduce evidence that controls for the demographic composition of early and same-day voters. Failure to do so leaves the primary alternative path (demographic sorting) unblocked, and the statistical deviation is therefore consistent with the null hypothesis of fair elections. This is the graph-theoretic counterpart of the standard statistical requirement to control for confounders.

Principle 3: Residual anomaly after demographic control triggers formal forensic audit. If, after controlling for demographics, a statistically significant and geographically systematic residual anomaly remains in the early vote distribution - one that cannot be attributed to known data-generating processes - then the court has grounds to order a formal forensic audit. The audit’s purpose is to test whether the manipulation path can be further specified (identifying actors, mechanisms, and timing) or whether the resid-

ual is better explained by measurement imprecision, aggregation artifacts, or unmodeled demographic factors.

These three principles operationalize the Korean Supreme Court’s existing evidentiary standard in graph-theoretic terms, providing a principled basis for courts worldwide to evaluate the growing volume of statistical electoral integrity claims that accompany every major election.

8 Conclusion

8.1 Summary of Contributions

This paper has proposed a framework for operationalizing legal proof standards through probabilistic causal DAGs grounded in Pearl’s structural causal models. The central formal contribution is the identification of *reasonable doubt* as a graph-theoretic property - the existence of an active, unblocked alternative causal path that the evidence record has not d-separated - and its grounding in the Probability of Necessity as a computable threshold.

Four results follow from this identification.

First, the *Reasonable Doubt Condition* (Proposition 1) provides a checkable criterion for structural adequacy of a verdict’s causal reasoning: PN on the prosecution’s path must meet the applicable threshold, and every active alternative path must be d-separated by the introduced evidence. The criterion is traceable to specific passages in the evidentiary record via the Weight Elicitation Protocol (Definition 8), making it reproducible and contestable by appellate review.

Second, the *Civil/Criminal Standard Duality* (Proposition 2) shows that the beyond-reasonable-doubt and preponderance-of-evidence standards correspond to structurally different computational tasks on the same underlying graph. Criminal proceedings impose a path-closure obligation on the prosecution: all alternative paths must be d-separated. Civil proceedings impose a path-weighting competition: both parties construct DAGs and the

adjudicator evaluates which causal account carries greater total evidentiary weight. The clear-and-convincing standard sits between them as an intermediate threshold on PN combined with a constraint on residual active path weight. These structural differences explain why the cognitive demands on factfinders differ between criminal and civil contexts in ways that have not previously been formally characterized.

Third, the graph-complexity argument of Section 4 establishes that formal graphs are necessary, not merely convenient, in complex causal adjudication. The cognitive limitations on informal d-separation judgment are well-documented; legal verdicts are presented in precisely the verbal format most susceptible to those limitations. The adoption of formal graphs is therefore not a technical preference but a structural requirement for reliably enforcing the reasonable doubt standard in complex cases.

Fourth, the electoral integrity framework of Section 7 demonstrates that the DAG approach generalizes beyond criminal law to any adjudicative setting where causal claims must be evaluated against competing alternatives. The South Korean Supreme Court’s evidentiary standard - requiring plaintiffs to specify the perpetrator, timing, and method of any alleged violation (Supreme Court of Korea, 2022, 2024) - is the verbal expression of the graph-theoretic requirement to name all nodes on an asserted causal path. The forensic and regression evidence of Kim (2026) (Kim, 2026) provides an empirical d-separation certificate for the demographic-sorting alternative path in four Korean electoral cycles, supplying the structural verification the Court’s standard requires but did not formally receive.

8.2 The Comparison with Prior Work

The framework extends Kadane and Schum (Kadane and Schum, 1996) by moving from the observational to the counterfactual layer of Pearl’s inferential hierarchy, and by replacing posterior comparison with d-separation as the criterion for closing alternative hypotheses. It extends Halpern (Halpern, 2016) by adding epistemic uncertainty over structural equations (the weight function), by mapping binary actual causation onto graduated proof standards,

and by introducing the verdict audit as a distinct operation from the determination of causation given known facts. It extends Pearl and Tian-Pearl (Pearl, 2009; Tian and Pearl, 2000) by applying their PN bounds to evidentiary edge weights rather than population data, and by connecting the PN threshold to the hierarchy of legal proof standards.

8.3 Limitations

Several limitations warrant acknowledgment.

The Weight Elicitation Protocol (Definition 8) requires human judgement in the assignment of epistemic scale ratings to evidentiary passages. Inter-rater reliability has not been empirically measured; a validation study with legal practitioners is needed before the protocol can be deployed in production appellate settings.

The PN bounds computed from edge weights are only as valid as the faithfulness of the weight elicitation to the verdict’s actual evidentiary structure. If important evidence is not recorded in the verdict text - as is common in jury trials where the factfinder’s reasoning is not required to be stated - the DAG may omit nodes or edges that would materially alter the path analysis. This is a limitation shared with all verdict-based audit approaches and is not specific to the present framework.

The electoral application is grounded in the Korean case, which has the advantage of a detailed published forensic and regression analysis (Kim, 2026). Generalization to other jurisdictions and electoral systems requires case-by-case assessment of which alternative causal paths are plausible and which forms of evidence constitute adequate d-separation in context.

The multiplicative path-weight approximation $P(H \mid do(\pi)) \approx \prod_e w(e)$ assumes path-level independence and is exact only for chain graphs with no shared intermediate nodes. When competing causal paths converge on a common mediator, the correct computation requires marginalizing over that node using Eq. (9), which in turn requires either a specified structural equation or additional parametric assumptions beyond the Weight Elicitation Protocol as currently defined. The case studies presented here are either unaffected by this

limitation (Willingham, Korean elections), trivially corrected by near-zero alternative path weight (Cadden), or handled by direct SCM reasoning rather than multiplication (Kennedy). The one genuinely affected case (Burrage) yields a robust qualitative finding because the shared node’s outgoing weight cancels. Future applications to cases with multiple shared mediators and non-negligible competing path weights should extend the protocol to elicit functional forms for each shared node’s structural equation. Nevertheless, this extension still does not require the population-level SCM analysis and structural equations only represent assumptions made in a trial, which can transparently be verified along with their causal conclusions.

8.4 Directions for Future Work

Four extensions are warranted.

First, the framework should be evaluated on a large-scale corpus of appellate decisions to characterize empirically the frequency and distribution of structural causal deficiencies across case types, jurisdictions, and time periods. Such an evaluation would also provide the inter-rater reliability data the Weight Elicitation Protocol requires.

Second, the civil/criminal duality of Proposition 2 should be extended to multi-party and conspiratorial causation structures, where the causal graph has multiple defendant nodes and the allocation of causal responsibility across parties is itself contested. Environmental and pharmaceutical liability litigation are natural test cases.

Third, the three procedural principles for electoral integrity adjudication (Section 7) should be assessed against the evidentiary rules of specific jurisdictions to identify where legislative or procedural reform would be needed to implement them. The Korean case provides a working example; comparative analysis with US, EU, and other constitutional systems would broaden the framework’s practical reach.

Fourth, automated extraction of causal graphs from verdict text - noted in Section 6 - requires annotated training corpora, calibration benchmarks, and inter-rater reliability

studies. The Weight Elicitation Protocol defines what such a system must produce; the development and evaluation of the system itself is left to future work.

8.5 The Broader Argument

The deeper argument of this paper is simple. When the causal structure of a case is complex enough that a formal graph is needed to track alternative hypotheses, the absence of such a graph is not a neutral omission. It is an invitation for the systematic cognitive errors that research on conditional independence judgment documents: mistaking a higher posterior for a closed path, mistaking a lower posterior for an adequate refutation, confusing mediation with confounding.

The reasonable doubt standard was designed to protect defendants from exactly these errors in the factfinder’s reasoning. A formal causal graph enforces the standard in the highest manner. By requiring that alternative causal paths be explicitly represented and formally closed - rather than verbally dismissed - the DAG framework ensures that “beyond a reasonable doubt” names a structural property of the verdict’s causal argument, not merely the confidence of the person who wrote it.

Institutional legitimacy as a systemic consideration. It is sometimes said that the law is not obliged to dispel every doubt, however irrational, and that a legal system cannot be held hostage to the unwillingness of any party to accept its rulings. This is formally correct.

Yet recent events suggest a more pragmatic perspective. In South Korea, unresolved public distrust of electoral procedures - distrust that no court formally validated, and that the forensic evidence examined here does not support - nonetheless accumulated sufficient political force to motivate an unconstitutional declaration of martial law, triggering a constitutional crisis whose consequences extended far beyond any single election dispute.

The lesson is not that courts must capitulate to unfounded suspicion, but that a legal system’s long-run authority depends partly on its perceived capacity to *demonstrate*, not

merely assert, that it has examined and closed the relevant alternative explanations. A formal evidentiary framework that makes the closure of causal paths explicit, auditable, and publicly legible is therefore not only an instrument of individual justice. It is a contribution to the institutional credibility on which democratic legitimacy rests.

The DAG framework, by rendering causal reasoning transparent and contestable, offers courts a means to show their work in precisely the settings where public trust is most fragile.

Data Availability and Declaration of Interests

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